

PhishDriftBench + DASH

The Reported–Deployed Gap in Phishing URL Detection: A Time-, Source-, Provenance- and Evasion-Aware Benchmark with Drift-Aware Selective Hardening

Final Year Project — Project Definition, Literature Review and Architecture

1. Abstract

Published phishing URL detectors routinely report 97–99.5% accuracy, yet these figures are almost always produced under one evaluation protocol: a random or k -fold split of a single dataset, balanced at roughly 50% phishing prevalence, with no adversary in the loop. Each of those choices is individually defensible and collectively misleading. Random splits leak time, because phishing URLs arrive in campaigns whose near-identical members land in both train and test partitions. Single-source evaluation hides source-specific artifacts — a feature can carry *opposite* meaning in different corpora. Balanced test sets conceal the base-rate problem: a detector with 99% accuracy and a 1% false-positive rate, deployed at a real prevalence of $1:10^4$, yields precision near 1%.

This project builds **PhishDriftBench**, a four-axis evaluation protocol that measures detectors across time, across data sources, against rule-based and generative evasion, and at deployment-realistic class priors. Under this protocol we re-evaluate five representative architectures drawn from the base papers. We add a **feature-provenance audit** that separates statically computable lexical features from those requiring a live network lookup, and quantifies how much reported accuracy is attributable to lookup artifacts rather than phishing semantics. Finally we propose **DASH** (Drift-Aware Selective Hardening), a lightweight mitigation combining drift-stability feature weighting, unsupervised drift detection, a bounded active-labelling budget, and a calibrated abstention band. The project is **URL-only**: no page is fetched, rendered, hosted or transmitted at any point.

Status: design and protocol complete; experiments not yet run. Every quantitative claim in the accompanying paper draft is an explicit unfilled placeholder. No results are reported until they are measured.

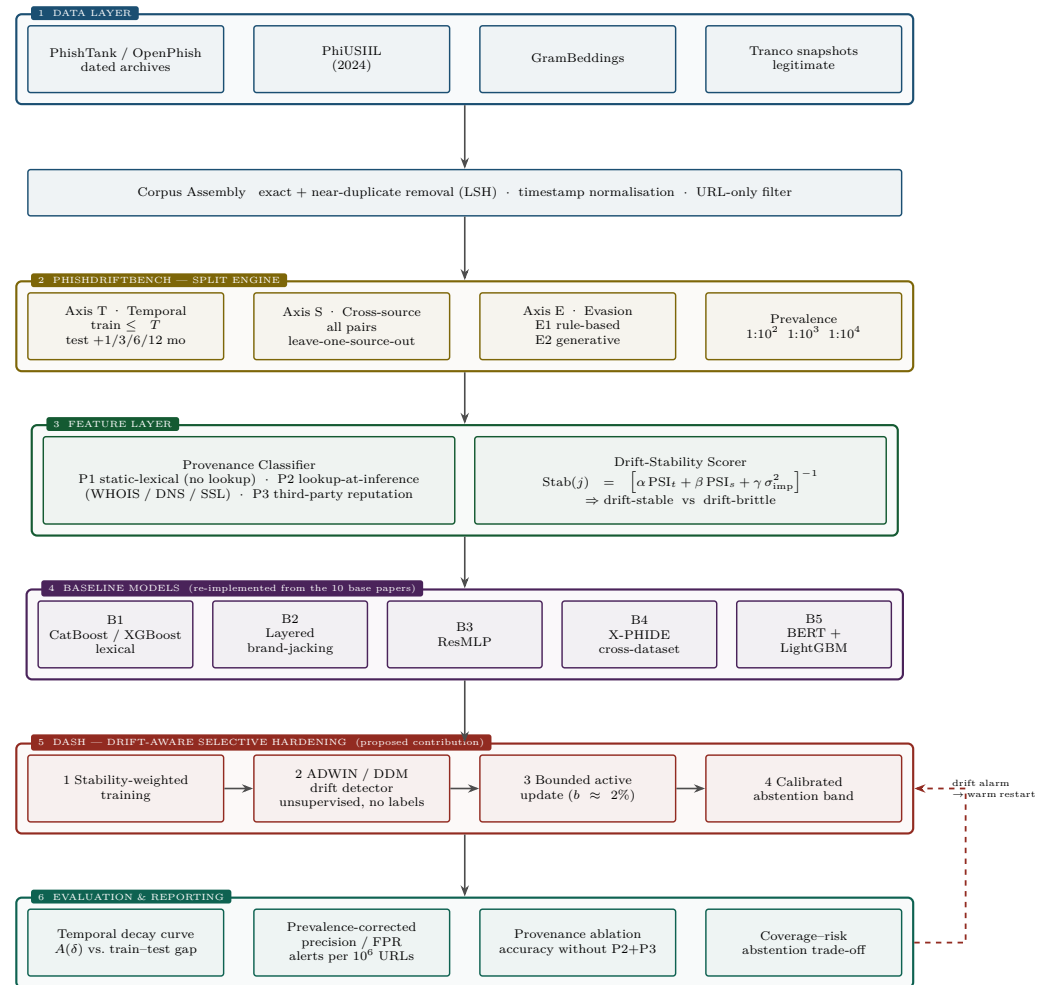
2. Improvements over the base papers

Each improvement names a specific, verified gap in the ten base papers.

- Temporal evaluation instead of random splits.** *Gap:* 0 of 10 base papers uses a chronological split. *Improvement:* train on URLs first observed $\leq T$, test on disjoint windows at +1, +3, +6 and +12 months; report a **decay curve**, not a single accuracy. Paper 5 accidentally demonstrates why — 99.35% on its 2025 data but 89.17% on an older corpus.
- True cross-source transfer.** *Gap:* only 1 of 10 (X-PHIDE) performs genuine transfer; Paper 5 evaluates on multiple datasets but **retrains per dataset**, which measures dataset difficulty, not transfer. *Improvement:* train once, evaluate on all source pairs plus leave-one-source-out.
- Prevalence-corrected reporting.** *Gap:* 0 of 10 reports precision at a realistic deployment base rate. *Improvement:* report precision, FPR and alerts per 10^6 URLs at prevalences of $1:10^2$, $1:10^3$ and $1:10^4$, alongside balanced figures for comparability.
- Evasion testing against our own detector.** *Gap:* Paper 10 catalogues cloaking thoroughly; **no method paper evaluates its own detector against that catalogue**. Paper 4 states outright that homograph attacks were never systematically evaluated. *Improvement:* **E1** label-preserving transformations from the cloaking taxonomy (homoglyph/IDN substitution, subdomain and path padding, percent-encoding, shortener wrapping, TLD swap, hyphenated brand insertion) and **E2** URLs from a contemporary generative model.

5. **Feature-provenance and lookup-dependency audit** (*most novel*). *Gap*: no base paper — and no adjacent work we could find — asks *where* a feature’s value comes from or *when* it was resolved. *Improvement*: classify every feature as **P1** static-lexical, **P2** lookup-at-inference (WHOIS, DNS, SSL, domain age, shortener expansion) or **P3** third-party reputation; then measure accuracy with P2/P3 ablated, accuracy when P2 features are resolved *today* versus at collection time, and recall when the lookup channel is unavailable or attacker-controlled. The hazard is concrete: Paper 6 reports WHOIS, DNS and domain-age features over a 549,346-URL corpus containing **only URL strings**. If those lookups were performed retrospectively they encode *domain mortality* — phishing domains die within days, legitimate ones persist for years — which any classifier can exploit without learning anything about phishing. Paper 4, which deliberately avoids third-party features, is the control.
6. **Drift-stability feature scoring**. *Gap*: drift is *named* by 4 of 10 papers and *implemented* by none. *Improvement*: score each feature by Population Stability Index across time windows and sources plus the variance of its permutation importance; partition into drift-stable and drift-brittle *before* deployment. Validation target: the HTTPS inversion X-PHIDE found (100% of legitimate URLs in PhiUSIIL, 59.2% in their own corpus) should be flagged without access to the second corpus’s labels.
7. **DASH — a mitigation, not just a diagnosis**. *Gap*: 0 of 10 proposes any drift mitigation. *Improvement*: (i) stability-weighted training; (ii) **unsupervised** drift detection over the prediction-confidence stream, requiring no labels — the binding constraint in deployment; (iii) a bounded active-labelling budget ($\approx 2\%$ of the window by uncertainty sampling) on drift alarm; (iv) a calibrated abstention band that escalates instead of guessing, reported as a coverage–risk curve.
8. **Duplicate-leakage probe (supporting)**. Report exact and near-duplicate rates across the train/test boundary for every corpus, and accuracy before and after deduplication. With k -fold CV over scraped corpora, campaign duplicates are near-certain.

3. System architecture



The pipeline is strictly URL-only. Layer 1 assembles dated corpora; Layer 2 generates the four evaluation axes; Layer 3 classifies features by provenance and scores their drift stability; Layer 4 hosts five baselines re-implemented from the base papers; Layer 5 is the proposed DASH mitigation; Layer 6 produces deployment-honest reporting. The dashed return path carries a drift alarm back into DASH for a warm restart.

4. Literature review

Ten base papers, all published 2023 or later: eight method or empirical studies plus two reviews.

#	Title	Authors	Publication & Year	Key Findings	Research Gap	Methodology	Future Work
1	An Effective Detection Approach for Phishing URL Using ResMLP	S. Remya; M. J. Pillai; K. K. Nair; S. R. Subbareddy; Y. Y. Cho	IEEE Access, vol. 12, 2024	98.29% accuracy; residual pipelining outperforms traditional ML baselines on the same features	Blacklists cannot identify dynamic and newly registered URLs; optimal detection accuracy remains an open pursuit	Lexical + sentiment + domain-age features → convolutional and inverted-residual blocks → MLP; Kaggle dataset, 80:20 random split	Address limitations affecting practical viability; explore self-organising networks and online-learning representations
2	On Phishing URL Detection Using Feature Extension	D. He; Z. Liu; X. Lv; S. Chan; M. Guizani	IEEE Internet of Things Journal, vol. 11, no. 24, 2024	Feature extension enriches information-poor URLs; two-layer BERT→CNN improves detection, including on cryptocurrency phishing	URLs are short and carry little signal; blacklist/whitelist methods have inherent limitations	TextRank keyword extraction builds a feature-extension library; BERT embeddings feed a CNN classifier; single real-phishing dataset	Extend to broader transaction-security threats in the blockchain and cryptocurrency ecosystem
3	Evaluating the Impact of Feature Engineering in Phishing URL Detection: A Comparative Study of URL, HTML, and Derived Features	Y. A. Kustiawan; K. I. Ghauth	IEEE Access, vol. 13, 2025	99.45% (CatBoost); derived features contribute most; systematic comparison across three feature families and ten models	Prior work studies URL or HTML features in isolation; no comparative study of engineered feature sets across models	101,063-URL corpus; URL / HTML / derived feature groups × 10 ML models; single dataset, random split	Cascading architecture — cheap URL analysis first, heavier content and relationship analysis only for borderline cases; craft further derived features
4	PhishOFE: A Novel Machine Learning Framework for Real-Time Phishing URL Detection With Optimized Feature Engineering	Y. A. Kustiawan; K. I. Ghauth	IEEE Access, vol. 13, 2025	99.48% (CatBoost) with no third-party features , enabling real-time deployment	Prior studies depend on outdated datasets and on third-party services that break in deployment	Optimised feature-engineering pipeline avoiding third-party lookups; CatBoost plus comparison models; single dataset, random split	Homograph-attack detection via character normalisation, visual-similarity metrics and Unicode-aware features — explicitly not evaluated in the paper
5	Enhanced Phishing Detection Approach Using a Layered Model: Domain Squatting and URL Obfuscation Identification and Lexical Feature-Based Classification	R. Goenka; M. Chawla; N. Tiwari	IEEE Access, vol. 13, 2025	99.35% (XGBoost) at 12.49 ms; but only 89.17% on an older dataset, which the authors attribute to 2025 target brands differing from 2018	Prevailing approaches overlook domain-squatting and URL obfuscation (brand-jacking), which cause most successful phishing	Layer 1: bad-domain features detect brand names in unwanted positions or misspelled forms; Layer 2: lexical-feature ML classifier; own corpus + 6 datasets, retrained per dataset	Extend the brand-jacking feature set beyond conventional lexical features
6	BGL-PhishNet: Phishing Website Detection Using Hybrid Model — BERT, GNN, and LightGBM	S. Remya; M. J. Pillai; B. S. Aparna; S. R. Subbareddy; Y. Y. Cho	IEEE Access, vol. 13, 2025	Hybrid 97.3% accuracy / 97.8% precision / F1 97.3 / ROC-AUC 0.97; individually BERT 90.4%, GNN 88.3%, LightGBM 85.8%	Single models lack flexibility, turnaround time and scalability on large, evolving datasets	BERT (semantic) + GNN (structural) + LightGBM (metadata) with ensemble late fusion; Kaggle corpus of 549,346 URLs; 10-fold cross-validation	Improve GNN memory efficiency; adapt the model to the evolving phishing landscape
7	Staying Ahead of Phishers: A Review of Recent Advances and Emerging Methodologies in Phishing Detection	S. Kavya; D. Sumathi	Artificial Intelligence Review, vol. 58, no. 50, 2025	Consolidated taxonomy spanning list-based, ML, DL, graph-based and GAN-based detection	No consolidated view of recent deep-learning, graph and generative approaches	Systematic literature review	Dataset diversity, adversarial robustness, interpretability and real-time deployability named as unfilled gaps
8	A Multimodal Phishing Website Detection System Using Explainable Artificial Intelligence Technologies	A. Vulfin; A. Sulavko; V. Vasiliev; A. Minko; A. Kirillova; A. Samotuga	Machine Learning and Knowledge Extraction (MAKE), 2026	F1 0.989 on own data, 0.953 on MTLP; quantifies an 8–15% temporal decay and a 3–9% adversarial drop; SOC deployment scenario on zero-day URLs	Single-modality detectors are brittle and offer no explanation to a human analyst	Late fusion of CatBoost (URL), 1D-CNN (characters), CodeBERT (HTML) and EfficientNet-B7 (screenshot); SHAP and Grad-CAM plus local LLM explanation	Improve explanation quality; broaden zero-day evaluation
9	XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation (X-PHIDE)	M. Misiek; T. Hyla	IEEE Access, vol. 14, 2026	97.80% single-dataset → 77.84% cross-dataset average (Custom 90.72, GramBeddings 70.65, PhiUSIIL 72.16); HTTPS <i>inverts</i> meaning across corpora	Dataset shift is “a factor often overlooked in existing literature”; single-dataset results do not transfer	XGBoost over a cross-dataset feature intersection; three distinct datasets; Chrome plug-in for real-time use	Operational challenges of distribution shift; model compression and lightweight deployment

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10	Uncovering the Cloak: A Systematic Review of Techniques Used to Conceal Phishing Websites	W. Li; S. Manickam; S. U. A. Laghari; Y.-W. Chong	IEEE Access, vol. 11, 2023	Taxonomy of server-side and client-side cloaking, 2012–2022; a small number of sophisticated campaigns account for over 89% of attacks	Cloaking and evasion techniques are scattered across the literature with no unified taxonomy	Systematic literature review (SLR)	Strategies targeting sophisticated campaigns; the role of certificate authorities; collaborative data sharing and faster takedown response

What the table shows at a glance

Evaluation property	Base papers satisfying it
Temporal / chronological split	0 / 10
Prevalence-corrected metrics	0 / 10
Genuine cross-source transfer	1 / 10 (X-PHIDE)
Evasion tested against own detector	0 / 10
Drift detection or mitigation implemented	0 / 10
Feature-provenance audit	0 / 10

Drift is *mentioned* by four papers and *implemented* by none.

5. Scope, feasibility and honesty constraints

URL-only. No crawling, rendering, hosting or third-party API calls. This sidesteps the dependency Papers 4 and 9 complain about, and makes the evasion axis safe: strings are perturbed or generated inside an offline dataset, never deployed.

Compute. Gradient boosting, a small char-CNN and one BERT fine-tune. A laptop suffices for most of it; requiring no GPU is itself a selling point for a real-time detector.

Riskiest ingredient: timestamps. Axis T needs first-observation dates. Where a source lacks them, dated dataset *releases* serve as a coarse time axis (2023 → 2024 → 2025 → 2026).

Non-negotiable. Every number in every results table must come from an experiment actually run. The paper draft carries explicit red `\resultTODO{}` placeholders wherever a measured value belongs. Fabricating results in work submitted for publication is research misconduct.

Novelty must be scoped honestly. Two concurrent works overlap parts of this proposal and must be cited: **PhreshPhish** (arXiv 2507.10854, 2025 — temporal splits, LSH leakage removal and base-rate benchmarks, but a preprint, website-based, and silent on feature provenance) and **Wei et al.** (IEEE Access 2025 — cross-dataset evaluation; found URL length alone yields 86.13% and 82.54% on two corpora). Claims in this project are stated as findings about *this corpus of ten papers*, not about the field as a whole.

Target venue. IEEE Access — seven of the ten base papers appear there, it publishes benchmark-plus-mitigation work of exactly this shape, and it is Scopus/SCIE indexed with a 4–8 week first decision. Fallbacks: IEEE ICC / GLOBECOM / TrustCom for a shorter conference version; Computers & Security for a slower, no-APC route.